

Response Letter with Annotations

Barberá, P., Casas, A., Nagler, J., Egan, P. J., Bonneau, R., Jost, J. T., & Tucker, J. A. (2019). Who leads? Who follows? Measuring issue attention and agenda setting by legislators and the mass public using social media data. *American Political Science Review*, 113(4), 883–901. <https://doi.org/10.1017/S0003055419000352>

Annotations

1. Purposes

- a. Motivation (one-sentence summary)
 1. The motivation to produce this study was to understand who leads whom in political agenda setting: do politicians follow public issue priorities or lead them?
- b. Specific Rationale (several bullet points)
 - Current literature demonstrates that who leads political agenda setting is unclear.
 - Furthermore, there is very little evidence surrounding which kind of citizens (general vs politically attentive vs. politician supporter) have the best probability to set political agendas among politicians
 - Previous survey data provided brief insights into the ways the public and political agendas covary, however, because of limitations with this data due to time and issue units available, it was not possible to show an association between public and politician agendas
 - However, in utilizing computational methods to increase granularity of measurements, researchers can begin to unpack who leads political agenda setting, politicians or the public?
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 1. IV:
 - a. Lagged issue attention of:
 - i. The same group (autoregressive terms)
 - ii. All other groups (cross lagged terms)
 2. DV:
 - a. Daily issue attention (log-odds of proportion of tweets about an issue) for each group
 3. Moderators:
 - a. Issue salience/relevance
 - b. Issue identity
 4. Mediator:
 - a. News media
 - Hypotheses:
 1. H1: The public's priorities predict the issues to which members of Congress subsequently pay attention.
 2. H2: Members of Congress initiate debates about issues that are subsequently followed by the public.

3. H3: Changes in attention allocation by the general public predict changes in issue attention by members of Congress.
4. H4: Changes in attention allocation by attentive publics predict allocation changes by members of Congress.
5. H5: Changes in attention allocation by party supporters predict allocation changes by members of Congress.
6. H6: To the extent that particular issues are more salient among the general public (according to the Downsian model), attentive citizens (according to the Attentive model), and party supporters (according to the Supporter model), these publics' priorities will more strongly predict the politicians' agenda.

2. Method

a. Research Design

- This study is an observational, longitudinal Twitter-based agenda-setting analysis, using a Latent Dirichlet Allocation (LDA) model (an unsupervised topic model) for issue coding and VAR/IRFs to estimate lead-lag relationships between legislators, public audiences, and media.

b. Sample (representative sample?)

- Twitter Sample of
 1. About 95% of legislators in the 113th Congress had active Twitter accounts, and all their original tweets (651,116) were included
 2. Random sample of US Twitter users
 - a. General Public
 - b. Attentive Public
 - c. Republican Supporters
 - d. Democratic Supporters
- This sample is only partially representative since it was collected via Twitter

c. Measures

- Populations measured
 1. Members of the 113th U.S. Congress (95% with Twitter accounts and 651,116 original total tweets).
 2. Four public groups: random U.S. Twitter users ("general public"), attentive public (follow ≥ 1 major news outlet), Republican supporters, Democratic supporters (follow $\geq$ co partisan Members of Congress, none from opposite party)
 3. 36 major U.S. media outlets.
- Core variables
 1. For each group, day, and issue: proportion of tweets about each political topic
 2. Political issues identified by Latent Dirichlet Allocation (LDA) on aggregated daily Members of Congress' tweets
 - a. 100 topics estimated; 53 judged political, then merged to 46 distinct political issues via human coding
 - i. Inter-coder agreement was 83%
 - ii. Cronbach's alpha was 0.92

d. Analysis

- Descriptive analyses
 1. Pearson correlations of issue attention distributions between Members of Congress', public groups, and media (issue attention congruence).
 2. Average daily attention per issue and group to define issue salience.
- Time series modeling
 1. For each group and issue, daily attention series transformed to log odds to handle skewed proportions.
 2. Vector autoregression (VAR) with 7 lags, including all groups plus media; issue fixed effects in main model, then 46 issue specific VARs
 3. Cumulative impulse response functions (IRFs) over 15 days: effect of a one time or permanent 10 percentage point increase in attention by one group on others' subsequent attention

3. Results and Conclusions

a. Answers to RQs and Hypotheses

1. H1: Supported.
 - a. Citizen attention generally predicts later congressional attention, and effects are larger than Members of Congress
2. H2: Partly supported
 - a. Members of Congress do lead supporters and attentive publics on some issues, but effect sizes are systematically smaller than public
3. H3: Not supported.
 - a. General public has the smallest and often insignificant effects on Members of Congress agendas, even in election year, and no evidence of issue-relevance amplification.
4. H4: Partially supported.
 - a. Attentive publics move Members of Congress, but less than partisan supporters, and issue-relevance correlations are small or absent, especially for Republicans
5. H5: Strongly supported.
 - a. Own-party supporters are the single strongest predictors of Members of Congress attention in VAR and impulse-response analyses
6. H6: Partially supported
 - a. Salience strengthens supporter publics→ Members of Congress effects (e.g., ACA, gun violence, IRS), but not for attentive or general publics.

b. Include Key Figures/Tables

TABLE 1. Description of the Tweets in the Dataset					
Group	N	Avg	Min	Max	Tweets
House Republicans	238	1,215	70	8,857	267,311
House Democrats	207	1,177	113	5,993	222,491
Senate Republicans	46	1,532	73	6,627	67,412
Senate Democrats	56	1,616	150	10,736	87,307
Random sample	25k	465	1	8,926	11,316,396
Informed public	10k	948	100	5,861	9,487,382
Republican supporters	10k	1,091	100	8,804	10,911,813
Democratic supporters	10k	1,306	100	5,122	13,058,947
Media outlets	36	7,803	8	15,858	273,121

Note: Period of analysis: January 1, 2013, to December 31, 2014. N corresponds to the number of Twitter accounts in each sample. Avg, Min, and Max correspond to the average, minimum, and maximum number of tweets, respectively, sent by individual users in each group during the whole period of analysis. Tweets corresponds to the total number of tweets sent by all users in each group during the period of analysis.

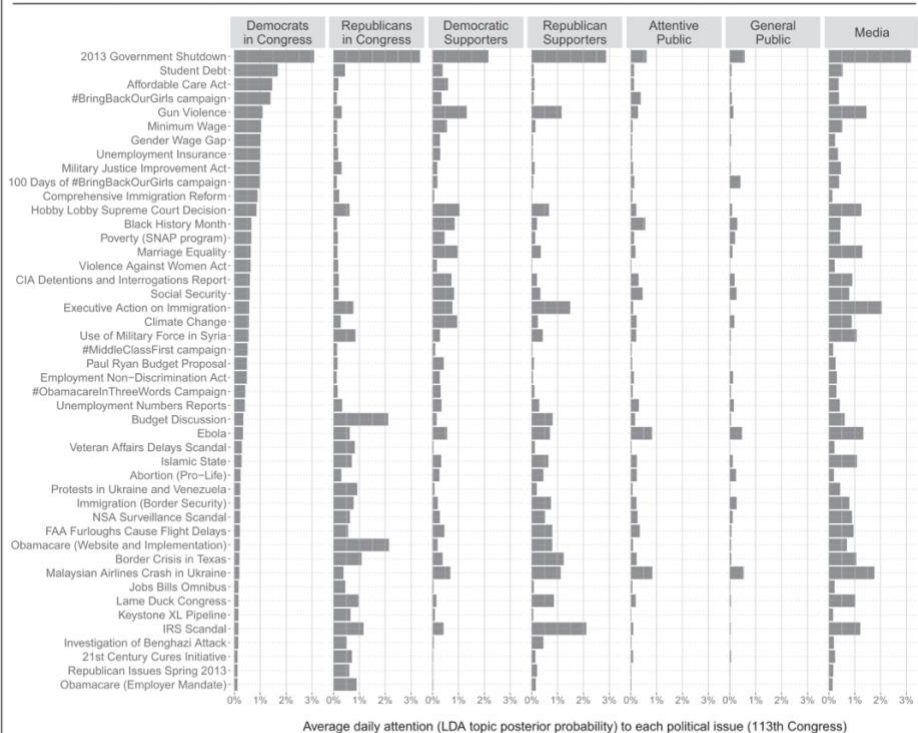
TABLE 2. List of Political Issues

Topic Number	Label	Topic number	Label
3	Investigation of Benghazi attack	50	Climate change
7	100 days of #BringBackOurGirls campaign	51	Lame duck congress
9	Gender wage gap	53	Minimum wage
12	Republican issues <i>Spring 2013</i>	58	Affordable Care Act
14	Marriage equality	62	Border crisis in Texas
15	Gun violence	63	Obamacare (employer mandate)
16	Abortion (pro-life)	64	FAA furloughs cause flight delays
18	Veteran affairs delays scandal	66	Malaysia Airlines crash in Ukraine
20	NSA surveillance scandal	67	Comprehensive immigration reform
23	#BringBackOurGirls campaign	70	#MiddleClassFirst campaign
28	Employment Non-Discrimination Act	75	Military Justice Improvement Act
32	Islamic state	81	Poverty (SNAP program)
33	Use of military force in Syria	83	Twenty-first century cures initiative
36	Ebola	85	Unemployment insurance
37	Social security	88	IRS scandal
39	Keystone XL pipeline	89	Obamacare (website and implementation)
41	Immigration (border security)	93	Jobs bills omnibus
43	Executive action on immigration	96	Violence Against Women Act
46	Unemployment numbers reports	97	Protests in Ukraine and Venezuela
47	Paul Ryan budget proposal	99	CIA detentions and interrogations report
48	Black history month	100	#ObamacareInThreeWords campaign
(101)	Student debt	(102)	Hobby lobby supreme court decision
(103)	Budget discussion	(104)	2013 government shutdown

Note: The topic number in parentheses indicate issues that have been created *ad hoc* by merging very similar topics from the topic model.

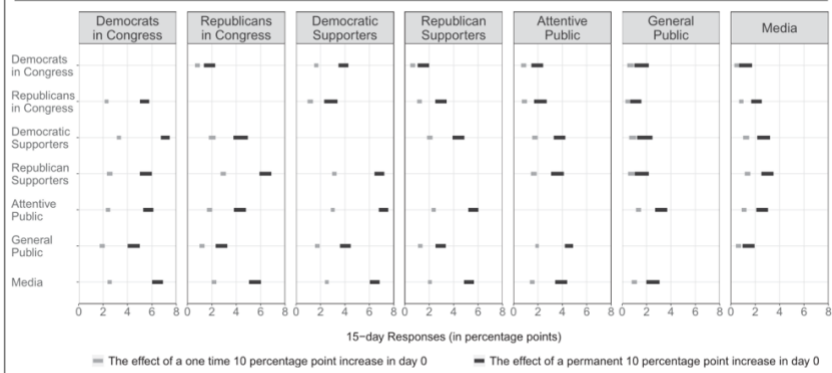
TABLE 3. Correlation in Issue Attention Between Members of Congress and Groups of the Public and the Media Over 46 Political Issues

Group	Democrats in Congress	Republicans in Congress
Democratic supporters	0.69	0.51
Republican supporters	0.41	0.77
Attentive public	0.49	0.52
General public	0.38	0.34
Media	0.52	0.63

FIGURE 1. Average Issue Attention by Groups of Politicians, the Public, and the Media

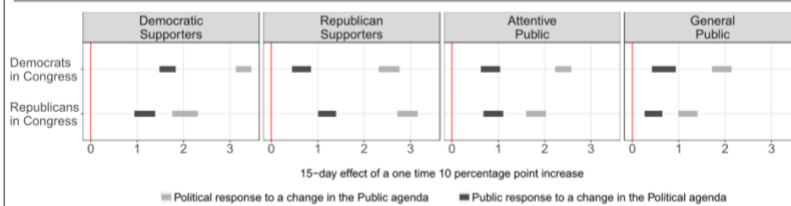
Note: Attention is represented as daily posterior LDA topic probabilities expressed in percentages. These are percentages based on all 100 topics of the LDA model.

FIGURE 2. 15 Day Cumulative IRFs: Predicted Issue Responsiveness Across Groups



Note: The coefficients (with 95% confidence intervals) indicate (in percentage points) how much more cumulative attention the groups in the panel titles paid to a given issue as a result of the groups in the y-axis increasing the attention to the same issue by 10 percentage points once (in gray) and permanently (in black) 15 days ago.

FIGURE 3. Politicians' Ability to Set Public Agendas versus the Ability of the Public to Influence Political Agendas



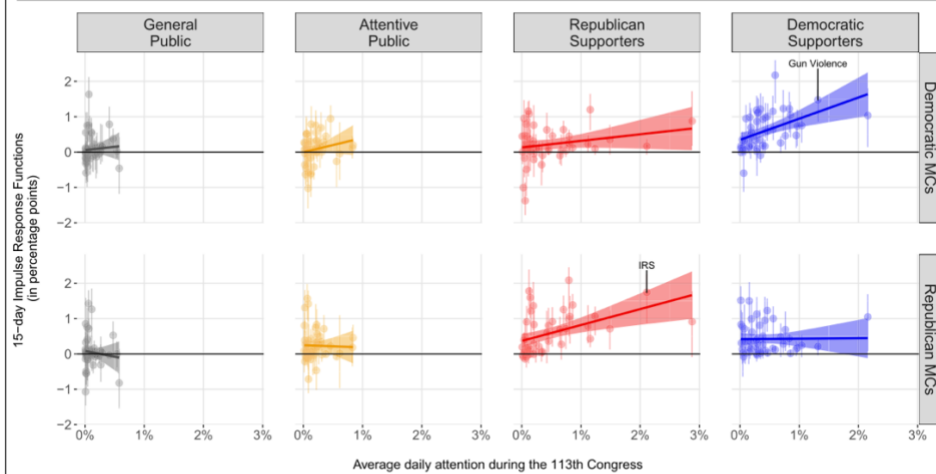
Note: The coefficients (with 95% confidence intervals) indicate (in percentage points) how much more cumulative attention the groups in the panel titles paid to a given issue as a result of the groups in the y-axis increasing the attention to the same issue by 10 percentage points (in black) 15 days ago. The gray coefficients indicate the vice versa effect.

FIGURE 4. Predicted Issue Responsiveness Across Issues and Groups (15 Day IRFs)



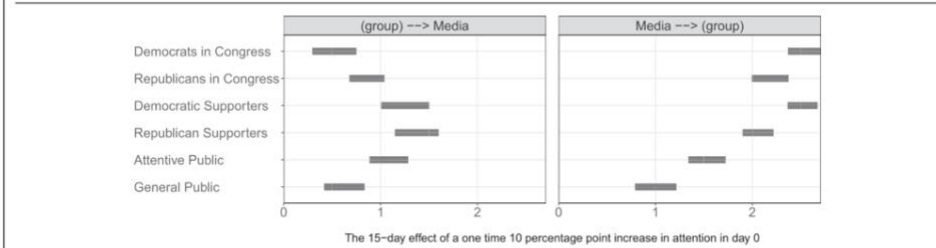
Note: The coefficients (with 95% confidence intervals) indicate (in percentage points) how much more/less cumulative attention the groups in the panel titles paid to the issue in the y-axis as a result of a group (identified by the color) increasing the cumulative attention to the same issue by 10 percentage points 15 days ago. Only coefficients not crossing zero have been included. The two left-most panels show the influence of the public on Members of Congress. The four right-most panels show the influence of Democratic and Republican members of Congress on the public. Versions of this figure that also show the coefficients crossing zero are available in Online Appendix C.

FIGURE 5. Correlation Between Public Issue Relevance and the Ability of the Public to Set Political Agendas



Note: The x-axis indicates the average attention the groups in the top panel titles paid to each political topic during the 113th Congress. The y-axis indicates how much more/less cumulative attention Democrats (top four panels) and Republicans in Congress (bottom four panels) paid to these topics as a result of the groups in the top panel titles increasing the attention to the topic by 10 percentage points 15 days ago. Each dot represents a different political issue and the lines around the dots represent 95% confidence intervals. Rows are sorted by the largest effect of Democrats in Congress (left panel).

FIGURE 6. Predicted Media Effects



Note: The effects (with 95%) in the left panel indicate how much the media outlets increased their attention to a given issue (in percentage points) 15 days after the groups in the y-axis increased their attention to the same issue by 10 percentage points. The coefficients in the right panel indicate the vice versa effects, how much the groups in the y-axis increase their attention to an issue 15 days after the media increased the attention to the same issue by 10 percentage points.

TABLE 4. Correlation in Issue Attention Between Media Outlets and the Other Groups of Analysis

Group	Media
Democrats in Congress	0.52
Republicans in Congress	0.63
Attentive public	0.74
Democratic supporters	0.79
Republican supporters	0.79
General public	0.55

c. Authors' Conclusions

- This study found that legislators more often follow than lead public attention on issues; public-to-elite effects are larger than elite-to-public effects. Furthermore, the strongest influence on Members of Congress comes from their own party supporters, then attentive publics; however, there are very weak effects from the general public. Media both lead legislators and publics and are especially responsive to partisan publics, which further amplifies party supporter influence. This relationship between the media and legislators is stronger among Republican legislators compared to Democratic legislators. Additionally, responsiveness among Members of Congress is strongest on issues highly salient to a party's own supporters which further reinforces polarization and inequality. Finally, media behavior further privileges partisan agendas and is partially to blame for inequality in issue responsiveness.

4. Discussion

a. Implications

- Important implications surrounding democratic politics as paying more attention to issues salient towards politically attentive audiences and party supporters can exacerbate political polarization.
- This work finally answers a long-standing open-ended question about who is leading political agendas. From this work we know it is led by politically involved publics and legislators follow.

b. Limitations

- Observational VAR design cannot fully rule out confounding; results are descriptive, not strictly causal
- Focus on Congress and Twitter users doesn't adequately represent the entire U.S. population. This study does not account for citizens that are non-Twitter-users and offline publics
- This study assumes a stable topic structure and excludes the Presidential agenda

c. Future Directions

- Future studies should test similar dynamics for the President, state-level politics, and across countries with different competition levels
- Examine how effects presented in this study are conditional based on issue type
 1. Here researchers can test the relationship between issue responsiveness and issue ownership.

--- Page Break Here ---
Response Letter (~600 words)

- Your assessment of the study

Overall, I thought this study was incredibly comprehensive and a fascinating way to assess agenda-setting effects outside of a strictly media context. To my knowledge the majority of studies that assess and use agenda-setting theory are done through a content analysis or survey methods. However, as this study addressed, previous work attempting to unpack the relationship between the politicians' and the public's agendas using these methods were insufficient in their analysis due to data limitations, like time and issue units available, and lack of granularity. In utilizing an observational analysis with a Latent Dirichlet Allocation (LDA) unsupervised model, authors were able to empirically evaluate who is setting political agendas. This was incredibly innovative for this field of study and opened up new ways to assess agenda setting.

Furthermore, this study expanded my view of agenda setting theory as I was initially taught how agenda setting is primarily understood as the media setting the agenda for the public. This was an interesting shift in agenda-settings theoretical application as media is still involved (Twitter) however, within this study the public or politician's tweets are drivers to predict political agendas and subsequently policy responsiveness. This felt innovative as a theoretical application, however, this could be due to a difference in academic field. At a glance, the majority of studies addressing agenda setting in this study come from a political science background which may bring a difference in conceptual understanding of agenda setting. Nonetheless this was an incredibly interesting application to understand agenda setting.

Lastly, I thought the methodology behind this study was well thought out; the authors provided well-rounded explanations to the reader throughout, and each methodological choice was further justified. The detail especially surrounding the Citizen data selection was incredibly detailed. Unpacking how each group was defined, how they gathered the samples, what additional data was pulled, and how they pulled it was incredibly exhaustive. Moreover, the use of a seven-lag structure provided an effective amount of time to determine which groups led conversations on political topics. Additionally, the authors' thorough explanation of the seven-lag structure accounting and controlling for the autocorrelation and partial autocorrelation functions was especially helpful in my knowledge building surrounding computational methods and its components.

- Weaknesses

While this study was well-rationalized and used innovative methods to assess the relationship between citizens and politicians setting political agendas, there are weaknesses that are important to acknowledge. One of the larger weaknesses of this study was how this only accounts for agenda setting on Twitter. While this study states that their goal was "not to assess whether a social media platform such as Twitter is a useful agenda setting tool for politicians or the public but rather to use their "tweets" as a proxy to measure attention being paid to political issues" (p.884), they are fundamentally only assessing how agenda setting works on social media platforms like Twitter. Their findings only apply within the Twitter space. In order to understand the broader political environment and whether the public vs. politicians drive political agendas we need move beyond social media and implement other methodologies.